



Generative Clinical Narrative and Risk Explanation from ICU Time-Series Data Using Large Language Models



M.Tech Final Semester Project (4th Sem)
Using eICU Collaborative Research
Database



Objectives

- Transform structured ICU data into natural language summaries
 - Apply Large Language Models for clinical narrative generation
 - Improve interpretability and clinical decision support

Architecture Overview

- User → Streamlit UI → Page Handlers
- Modules: eicu_eda, eicu_data, eicu_features
- Database: SQLite eICU
- Supports modular data processing via agents

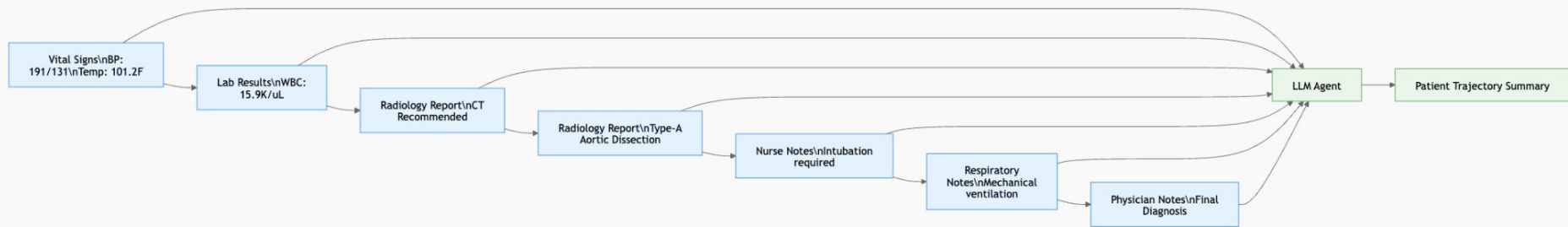
Data Set (eICU Collaborative Research Database) & Research Papers

- <https://physionet.org/content/eicu-crd-demo>
- https://physionet.org/login/?next=/projects/eicu-crd-demo/2.0.1/request_access/4 (Get Access to Complete Data)

Research Papers

- Improving Representation Learning of Complex Critical Care Data with ICU-BERT <https://arxiv.org/pdf/2502.19593>
- Zero-Shot Large Language Models for Long Clinical Text Summarization with Temporal Reasoning
<https://arxiv.org/pdf/2501.18724v2>

Agent Workflow



Navigation Flow

- Sidebar controls navigation
- Pages: Summary, Global Overview, Time Series
- Patient Report, Cohort Features, Table Preview
- Each page calls specific handlers

Patient Data Flow

- Select patient unit stay id
- Load patient-related tables
- Build patient data bundle
- Generate reports dynamically

Report Generation Flow

- User selects report type
- System routes to render functions
- Generates ICU Nurse, Ventilator, Lab reports
- Uses unified data bundle

AI Integration (in-progress)

1. BeeAI agent workflow integration
2. OpenAI LLM for insights
3. Natural language query support
4. Agent Stack for deployment

Code Base

- <https://github.com/suryashekhawat/MTech4SEM>

Thank You